

Autonomous Map Lead Scraper & **Multi-Channel Outreach Engine**

Complete Technical Architecture, Operational Blueprint, Anti-Ban Resilience & International Client Acquisition Engine (UK, UAE & Global).

 Google Places API (New) Real Scraping

 Deep Website Crawler & Email Extractor

 WhatsApp Automation (Typing Emulation)

 Telegram UserBot (GramJS Resolution)

 Gmail API 1-Click Draft Generation

 SQLite WAL High-Concurrency CRM Engine

Engineering & Founders:

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AI AutomationHubs India
ai-automation-hubs.com • September 2026
Confidential System Documentation

Executive Summary & System Overview

AI AutomationHubs LeadFlow Pro is an autonomous, full-stack B2B lead generation and customer outreach engine engineered for rapid client acquisition across international commercial hubs (United Kingdom, UAE, USA, and India).

Key Operational Metric: Eliminates manual prospecting by automatically mining authentic businesses from Google Maps, crawling official websites for direct emails and decision-maker phone numbers, and initiating outreach across WhatsApp, Telegram, and Gmail with zero ban risk.

Core Capabilities

**Real Places Mining**
Direct integration with Google Places API v1 (New) ensuring 100% verified physical addresses, verified ratings, phone numbers, and official websites. Zero mock/fake records.

**Deep Website Enrichment**
Crawls business homepages, contact-us, and about pages to extract official domain emails, WhatsApp click-to-chat links, and social links (Instagram, LinkedIn).

**Multi-Channel Outreach**
Integrated with WhatsApp Web (Puppeteer), Telegram UserBot (MTPROTO GramJS), and Gmail API for personalized 1-click & automated outreach.

Technology Stack Specifications

Layer	Technologies / Libraries	Architectural Role & Purpose
Backend Runtime	Node.js v24.19 (ESM Mode)	Asynchronous event-driven server runtime handling long-polling and scraping tasks.
Database	better-sqlite3 v13.0	Embedded SQL database with Write-Ahead Logging (WAL) for high concurrency and zero locks.
Maps Data	Google Places API v1 (New)	Queries Google Maps search endpoint for global business listings with field masking.
WhatsApp Core	whatsapp-web.js (Pedroslopez Edge)	Puppeteer-controlled headless Chromium running WhatsApp Web with session caching.
Telegram UserBot	GramJS (telegram npm v2.26.22)	Direct MTPROTO connection to Telegram servers as authenticated user (Tanuj Chandel).
Email Outreach	googleapis v140.0.1	Official Google OAuth2 client for drafting personalized outreach in Gmail.
Phone Intelligence	libphonenumber-js v1.13	Validates worldwide numbers, detects landline vs mobile, and normalizes into E.164.

Lead Scraping & Phone Intelligence Engine

The data collection pipeline is designed with a strict **Zero Fake Data policy**. It operates in multiple cascading phases to guarantee authentic business names, verified contact info, and high delivery rates.

Phase 1: Google Places API (New) Extraction

When a search is launched (e.g. *"Private Dental Clinics in London, UK"* or *"Real Estate Agencies in Dubai, UAE"*), the engine calls Google's latest v1 Places API endpoint:

```
POST https://places.googleapis.com/v1/places:searchText
Headers:
  Content-Type: application/json
  X-Goog-API-Key: [API_KEY]
  X-Goog-FieldMask: places.displayName,places.formattedAddress,places.nationalPhoneNumber,
                    places.internationalPhoneNumber,places.websiteUri,places.rating,
                    places.id,nextPageToken
Body:
  { "textQuery": "Real Estate Agency in Dubai, UAE", "pageSize": 20 }
```

Pagination & Rate Control: If more than 20 leads are requested (e.g., 50 leads), the engine tracks `nextPageToken`, pauses for 1,500ms to allow Google token registration, and cycles through consecutive pages until the quota is satisfied.

Phase 2: Deep HTML Website Crawler

For every business that has an official website listed on Google Maps, the scraper runs an autonomous multi-page crawler across `/`, `/contact`, `/about`, and `/reach-us`.

- 1. Direct Business Emails:** Uses regular expressions combined with domain blacklists to capture authentic corporate inboxes.
- 2. WhatsApp Click-to-Chat Numbers:** Scans for `wa.me/<number>` and `api.whatsapp.com/send?phone=<number>` embedded in chat widgets. This captures direct mobile phones even when the Google Maps listing displays an office landline!
- 3. Social Proof Profiles:** Automatically pulls links to Instagram, LinkedIn company pages, Facebook, and Telegram channels.

Phase 3: International Phone Classification

Format Pattern	Region / Carrier	Classification	Delivery Channels
+971 50 / 52 / 54 / 55 / 56 / 58	UAE Mobile (Etisalat / du)	Mobile Ready	WhatsApp, Telegram, SMS, Call
+971 4 xxx xxxx	Dubai Landline (Fixed Line)	Fixed Office Desk	Office Direct Call, Email
+44 7xxx xxxxxx	UK Mobile	Mobile Ready	WhatsApp, Telegram, SMS
+44 20 xxxx xxxx	London Landline (Fixed Line)	Reception Desk	Direct Call, Verified Email
+91 6xxx - 9xxx	Indian Mobile (Jio/Airtel/Vi)	Mobile Ready	WhatsApp, Telegram, SMS

Autonomous Multi-Channel Outreach Systems

1. WhatsApp Automation Client

Underlying Engine: Headless Chromium via Puppeteer with WhatsApp Web interface.

- **Human Typing Emulation:** Triggers realistic typing presence for 600–1,400ms prior to dispatching messages.
- **LID & JID Pre-Verification:** Queries `client.getNumberId()` to confirm registration before sending.
- **Anti-Detection Stealth:** Modern desktop User-Agent, disabled `AutomationControlled` blink feature.
- **Auto-Recovery Watchdog:** Automatically clears stale locks (`DevToolsActivePort` , `SingletonLock`) and self-heals in 5s.

2. Telegram UserBot (MTPROTO GramJS)

Underlying Engine: GramJS MTPROTO client operating directly as Tanuj Chandel's authenticated account.

- **Phone-to-Entity Resolution:** Resolves cold business numbers through `Api.contacts.ImportContacts` before messaging.
- **Permanent SQLite Session:** Session string stored in SQLite settings; never requires re-authentication.
- **Anti-Flood Throttler:** Automatically spaces consecutive messages with randomized 15–25s pauses.
- **1-Tap & Bulk Execution:** Accessible via individual row buttons and bulk operations bar.

3. Gmail API OAuth2 Engine

Underlying Engine: Google APIs Node.js Client with token persistence.

- **1-Click Draft Generation:** Creates fully personalized drafts directly in founder's Gmail inbox.
- **Dynamic Placeholders:** Injects `{{BusinessName}}` , `{{Location}}` , `{{Rating}}` .
- **Human-in-the-Loop Safety:** Drafts are created for review, giving 100% control over tone and timing.

4. Real-Time Telegram Founder Alert Bot

Underlying Engine: Telegram Bot API integrated with SSE event listeners.

- **Instant Inbound Alerts:** Whenever a client replies on WhatsApp or Telegram, an instant alert is pushed to founders.
- **OpenAI Suggested Reply:** Analyzes the inquiry and generates an AI draft response ready for 1-tap dispatch.
- **1-Tap Quick Action Links:** Direct `wa.me` deep links and `tel:` links included in every notification.

International Market Case Studies & Live Data

The system has been verified across two primary international commercial hubs: **London, United Kingdom** (Private Healthcare & Dental) and **Dubai, United Arab Emirates** (Real Estate & Property Brokers).

Campaign A: 50 Real Estate Agencies in Dubai, UAE

Market Profile: Dubai property brokers have extremely high transaction sizes (AED 2M – 50M). Missing an inquiry costs thousands of dollars, making them prime buyers for 24/7 AI Lead Qualification Bots.

Agency Name	Contact Number	Official Business Email	Rating	Primary Outreach
Time Homes Real Estate	+971 56 448 4762 (Mobile)	info@timehomesrealestate.com	4.7 ★	WhatsApp + TG
Axis Real Estate Brokerage	+971 52 875 9269 (Mobile)	info@axisrealestate.com	4.9 ★	WhatsApp + TG
Impressive Real Estate LLC	+971 50 746 5012 (Mobile)	shetty@impressiverealestate.net	4.8 ★	WhatsApp + TG
TheRealtorDubai Brokerage	+971 55 195 2518 (Mobile)	info@therealtordubai.com	5.0 ★	WhatsApp + TG
Homeland Realty Real Estate	+971 55 661 0000 (Mobile)	info@homeland.ae	4.9 ★	WhatsApp + TG
Luxfolio Real Estate	+971 4 335 6007 (Office)	info@luxfoliorealestate.ae	4.6 ★	Direct Email

Campaign B: 50 Private Dental Clinics in London, UK

Market Profile: UK practices list reception desk landlines (+44 20 ...) on Google Maps and operate almost entirely on corporate email. 36 verified direct emails were extracted, and 36 Gmail drafts were generated in 1-click.

Clinic Name	London Desk Phone	Verified Clinic Email	Rating	Outreach Channel
Pro Dental Clinic	+44 20 3026 4402	enquiries@prodentalclinic.london	4.7 ★	Gmail Drafted
Elleven Dental London	+44 20 7487 2711	info@ellevendental.com	4.8 ★	Gmail Drafted
Square Mile Dental Centre	+44 20 7377 0990	info@squaremiledentalcentre.co.uk	5.0 ★	Gmail Drafted
Dental Care London	+44 20 3538 9449	smile@dentalcarelondon.co.uk	4.8 ★	Gmail Drafted
Marylebone Smile Clinic	+44 20 7126 8526	info@marylebonesmileclinic.co.uk	5.0 ★	Gmail Drafted

Database Architecture, Endpoints & Operations

SQLite Database Schema

📄 leads Table Structure

```
id TEXT PRIMARY KEY,
createdAt TEXT,
searchTerm TEXT,
location TEXT,
businessName TEXT,
address TEXT,
phone TEXT,
website TEXT,
email TEXT,
rating REAL,
segment TEXT,
qualityScore TEXT, -- Hot, Warm, Cold
leadStatus TEXT, -- New, Contacted, Interested, Closed
emailStatus TEXT, -- Pending, Draft Created, Sent
whatsappStatus TEXT, -- Pending, Sent, Error
telegram TEXT, -- Sent, Not on Telegram
notes TEXT,
dataMode TEXT -- REAL
```

⚙️ locations & settings Tables

```
-- locations table (Scrape Jobs)
id TEXT PRIMARY KEY,
term TEXT,
location TEXT,
maxLeads INTEGER DEFAULT 20,
status TEXT, -- Pending, Scraping, Done
error TEXT,
createdAt TEXT

-- settings table (Stored Credentials)
id INTEGER PRIMARY KEY CHECK (id = 1),
data TEXT -- JSON containing API keys,
-- templates, Telegram session,
-- and OAuth tokens
```

Core API Endpoints Reference

Endpoint	Method	Description & Payload
/api/locations	POST	Creates a new scraping location. Body: { term, location, maxLeads }.
/api/locations/:id/run	POST	Triggers background scraper run. Body: { overwrite: boolean }.
/api/leads	GET	Returns filtered leads list with search, segment, and status queries.
/api/whatsapp/status	GET	Returns WhatsApp client link status (Disconnected , QR_Ready , Connected).
/api/whatsapp/disconnect	POST	Safely closes session, cleans stale lock files, and prepares fresh client.
/api/outreach/telegram	POST	Triggers bulk Telegram outreach to selected lead IDs with anti-spam delays.
/api/telegram-user/status	GET	Checks Telegram UserBot session state (configured, hasSession, isConnected).
/api/gmail/draft/:leadId	POST	Creates a personalized outreach email draft directly in Gmail inbox.
/api/logs/stream	GET	Server-Sent Events (SSE) stream broadcasting real-time logs to the UI.